Amit Chandran

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Mechanical Engineer with skills and experience in development of software, hardware and mechanical components within cross-functional teams. Aspiring to work within robotics, control systems or simulation.

Education

UCL - MEng Mechanical Engineering *Expected June 2028*

- Relevant Modules: Thermodynamics, Dynamics, Mathematical Modelling, Design, Materials
- Grade: Y1 – 85%

Experience

Systems Engineering Intern – Leonardo UK *June – August 2025*

- Implemented HDL-compatible signal processing pipelines in Simulink and deployed them on an RFSoC FPGA board, producing high-resolution spectra
- Developed a compression algorithm reducing FFT data bandwidth requirements by over **2500×**, minimising network utilisation whilst preserving critical information
- Directed the integration of upstream and downstream hardware/software sub-systems, delivering a coherent high-performance system
- Coordinated with software engineers to develop a robust data exchange protocol using TCP, enabling results to be viewed through a web-based server
- Architected a self-test algorithm to verify RF signal integrity, cutting downtime and boosting diagnostics
- Optimised max-hold algorithms using MATLAB to decrease run time by **99.3%**, reducing system latency

Control Engineer – UCL Rover Team *January 2025 – present*

- Engineered control software using Arduino IDE to remotely adjust the yaw and thrust of the rover
- Modelled and manufactured enclosures for ESCs (Electronic Speed Controllers) in Fusion 360, ensuring seamless integration into the mechanical structure of the rover
- Diagnosed and resolved errors within the program through a robust verification method using oscilloscopes, streamlining the troubleshooting process

Projects

IMechE Design Challenge – Autonomous Robotic Device *January – March 2025*

- Led the team to develop a bespoke electro-mechanical solution involving H-bridges and latch circuits, maximising autonomy over competitors and placing 3rd of 45 teams
- Designed and assembled a PCB using EasyEDA, cutting weight by **85%** as well as reducing internal wire length by **1.2** metres versus breadboard designs
- Formulated a rigorous test strategy and built a test rig to rapidly implement upgrades to the vehicle
- Collaborated with mechanical designers to prioritise DFMA (Design for Manufacturing and Assembly), maximising space efficiency and vehicle modularity

Industrial Cadets Gold Award - Digital Locker and Kiosk *October 2022 – April 2023*

- Modelled kiosk components within Fusion 360, focusing on enhancing modularity and **DFMA**
- Managed the integration of Arduino firmware with a Python application using PySerial, upgrading the manual lock and paper sign in with a digital system
- Organised the live demonstration of the product by ensuring cross-functional sub-teams were briefed on all components, leading to the judges awarding us the **Best Teamwork Award**

Skills

- **Programming Languages** – MATLAB, Arduino, Python, C++
- **Tools / Software** – Simulink, Fusion 360, Git, Azure DevOps, EasyEDA
- **Manufacturing** – 3D Printing, Laser Cutting, Standard workshop equipment, Soldering